TIA CURRENT DESIGN CHOICES AND CALCULATION BASIS

Architecture Specification: Hardware Implementation v1.0

System: Analog 2T1C Compute Matrix for Inverted MAML
Components: MCP6024 (Quad Op-amp), DOX3134A (Multiplier)
Status: Finalized architecture for Printed Circuit Board (PCB)
Measurement Window: 2 µs

1. System Framework and Constraints

To achieve a deterministic and linear analog computation within the matrix's rapid measurement window, the system's base levels are defined as follows:

- **Virtual Ground (Source, V_s):** `1.65 V` (Controlled via TIA).
- **Row Voltage / Data (Drain, V_d):** Swings from 1.65 V (Data 0) to a maximum of `1.90 V` (Data 255). This results in a maximum voltage drop V_{ds} = 0.25 V.
- **Weight Voltage (Gate, V_g):** Swings from 2.1 V (Weight 0) to a maximum of `3.1 V` (Weight 255).
- **Transistor Operating Region:** The voltages above guarantee that the DOX3134A operates strictly within the linear triode region.

Hard Limits for MCP6024 (Quad Op-amp):

- Absolute maximum short-circuit current is ±30 mA. This value must never be approached.
- According to the datasheet, the output saturates (clips) and loses its linear Rail-to-Rail capability at high sink currents (e.g., at 10 mA, it bottoms out at approx. 250–400 mV above ground).
- High currents through a single channel cause localized heating (thermal cross-talk) within the Quad package, generating dynamic drift in the input offset voltage (V_{OS}).

2. Trade-offs and Decisions Regarding Current Budget

Discussed Alternative: 10 mA max current per column.
Decision: Rejected. Allowing the TIA stage to sink 10 mA and the row Op-amp to source 7.5 mA introduces non-linearity, loss of Rail-to-Rail capacity, and unacceptable thermal drift.

Finalized Alternative: Max 5 mA per column

This provides a robust and safe current budget for an 8x6 matrix:

- **Current per cell:** At full scale (Data 255, Weight 255) across all 8 rows, each cell averages **625 μA** (5 mA / 8). This sits comfortably within the transistor's optimal operating zone.
- **Load on Row Op-amp (Source):** Each row Op-amp drives 6 columns. In the worst-case scenario, the row current becomes: 6 × 625 μA = **3.75 mA**. The MCP6024 handles this with excellent linearity at a potential of 1.9 V.

3. Dimensioning the TIA (Transimpedance Stage)

Because the current is driven from the row (1.9 V) into the virtual ground (1.65 V), the TIA output swings towards 0 V to balance the current across the feedback resistor (R_{rf}).

Selection of TIA Resistor (R_{rf}): 100 Ω

- **Output voltage at idle (0 mA):** 1.65 V
- **Output voltage at absolute max load (5 mA):** V_{out_TIA} = 1.65 V - (5 mA × 100 Ω) = `1.15 V`

Justification: Stopping at 1.15 V ensures a full 1.15 V margin to the bottom rail (0 V), keeping the Op-amp 100% linear within this window. A low R_{rf} also minimizes the Miller effect and maintains an extremely low-impedance column node, ensuring parasitic capacitances charge fully within the 2 μs measurement window.

4. ADC Adaptation (Secondary Op-amp Stage)

Using 100 Ω yields superior analog stability but compresses the active signal into a narrow voltage window (0.5 V swing). To preserve bit resolution during digitization, a buffering second stage is implemented.

- **Design Choice:** A difference amplifier built on a spare MCP6024 channel is placed after the TIA stage. This stage subtracts the baseline (1.65 V) and applies a specific gain.
- **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.
  $V_{ADC} = 6.6 \times (1.65\ V - V_{out\_TIA})$
- **Result:** At 0 mA output from the matrix, the ADC receives 0 V. At 5 mA, the ADC receives 3.3 V. High-frequency internal matrix noise is effectively isolated.

5. Physical Hardware Deviations and Inverted MAML

The system is designed on the principle that hardware imperfections are acceptable, as the Inverted MAML algorithm calibrates them out in the software layer.

Analysis of Voltage Drop (IR-Drop):

- Copper trace physics: Width 0.3 mm, thickness 35 μm (1 oz), max length approx. 6 cm.
- Calculated trace resistance: ≈ 0.1 Ω.
- Voltage drop at 5 mA max load: `0.5 mV` .

Compensation: A voltage drop of 0.5 mV corresponds to nearly 10% of an LSB, creating spatial asymmetry. The Inverted MAML algorithm registers this as a deterministic deviation during the calibration cycle and shifts the digital input and output weights in real-time to maintain mathematical perfection at the system level.